

RAMCO INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering
Academic Year: 2019 - 2020 (Even Semester)
INNOVATIVE PRACTICES

Degree, Semester & Branch: IV Semester B.E. CSE

Course Code & Title: CS8493 Operating Systems

Name of the Faculty member: Mrs. N. Nithya

Active Learning Strategy: Role Play

Topic: CPU Scheduling Algorithms

Date: 13.01.2020

Role play:

Role play encourages participation among passive learners, adds dynamism to the classroom and promotes the retention of material.

Benefit of the Role play:

- Students immediately apply content in a relevant, real world context.
- Students can transcend and think beyond the confines of the classroom setting.
- Students see the relevance of the content for handling real world situations.
- The instructor and students receive immediate feedback with regard to student understanding of the content.
- Students engage in higher order thinking and learn content in a deeper way.
- Instructors can create useful scenarios when setting the parameters of the role play when real scenarios or contexts might not be readily available.

Course Outcome:

CO2: The students will be able to apply various scheduling algorithms, deadlock prevention and deadlock avoidance algorithms.

Goal:

The students will be able to apply various scheduling algorithms.

Reason for choosing the particular topic (Method):

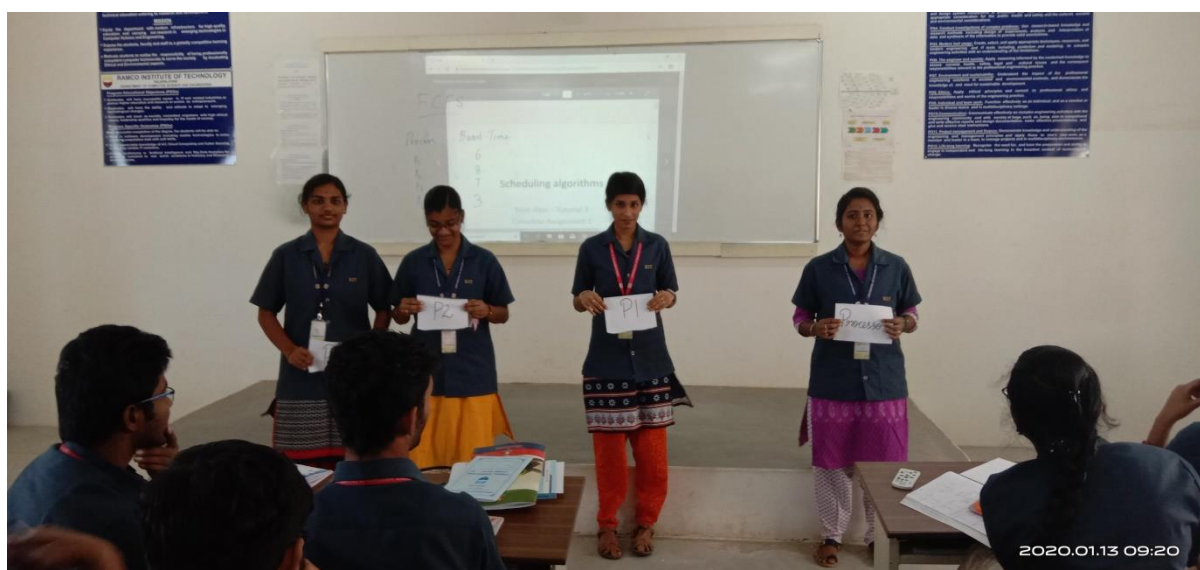
Students are asked to “act out” so they get a better idea of the concepts and theories being discussed. Role play helps the students to visualize the functioning of CPU scheduling algorithms. In addition, role play is used to empower, engage, and stimulate a classroom by putting students at the centre of the learning process.

How we implemented Role play:

- The faculty has discussed the concept of FCFS and SJF on the previous day and asked willingness from the students to role play the concepts on the next day.
- The students formed groups and prepared for the role play.
- The students enacted CPU scheduling algorithms like FCFS and SJF.
- Three processes with different burst time and arrival time are to be executed by the processor using FCFS and SJF.
- Students have taken the role of processes and processor.

Learning Outcomes:

The students were able to demonstrate various scheduling algorithms.



Reflective Critique:

Pre-implementation Reflection:

- Some students were not willing to participate which necessary for the execution of the role is playing activity.
- Less number of students was involved in the activity.

Post Implementation reflection:

- Students were able to identify and apply the working of scheduling algorithm
- All the students enjoyed the activity.
- Students' feedback reflected that they have understood the concept.
- A scheduler can be added in the next role play in addition to the processor and processes to depict closer to the real scenario in CPU scheduling algorithms.

Learning Outcomes/ Program Outcomes	PO1	PO2	PO3	PO9	PO10	PO12
LO: The students will be able to apply various scheduling algorithms.	3	3	1	2	1	2

References:

- <https://serc.carleton.edu/introgeo/roleplaying/whatis.html>
<https://www.ritrjpm.ac.in/images/computer-science/3-CS6701-CNS.pdf>
https://www.ritrjpm.ac.in/images/computer-science/10_CS8591_Rolepaly.pdf

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